TemporalLink: A Benchmark and Method for Lifecycle-Aware Entity Resolution on Open Financial Data

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ABSTRACT

We introduce TemporalLink, an open benchmark for *lifecycle-aware* entity resolution on financial entities — matching records across mergers, rebrands, spinoffs, and jurisdiction migrations, where the surface form has deliberately changed. The benchmark is built from public GLEIF Level-1/2, SEC EDGAR, and Wikidata: 125,519 labeled pairs over 3.3M entities, with three held-out evaluation regimes (time, event type, jurisdiction). Hard negatives are mined to be *name-similar* to each predecessor, and true successors are by construction *name-different* — so on this benchmark name-similarity baselines (token, char-TF-IDF, zero-shot E5) score below 0.5 AUROC on time and event splits. We emphasise that this is a property of how negatives were mined here, not a universal claim about name similarity in ER. A fine-tuned MiniLM cross-encoder (22M parameters, ≈17 min on MPS) is the first method we tested that escapes this regime, reaching 60.5 / 56.5 / 75.0% Hits@1 (±2.2 / 2.4 / 8.8 pp bootstrap CI), 24–35 pp over the best bi-encoder. We pair the headline number with a miner-robustness re-test that swaps E5-cosine for TF-IDF char-3gram mining: the cross-encoder advantage shrinks 5–15 pp under candidate-pool shift but stays +13 to +38 pp above zero-shot, evidence the gain is partly in-distribution but not purely a within-miner artifact. Per-event analysis foregrounds where the lift is real (M&A, absorption: +30–45 pp) and where it isn't (pure renames are already solved at zero-shot). Code, splits, predictions, and trained model weights reproduce end-to-end on a single MacBook in ∼2 hours.

KEYWORDS

entity resolution, record linkage, financial entities, lifecycle events, GLEIF, benchmark, cross-encoder

1 INTRODUCTION

Twitter Inc. became X Corp. Facebook became Meta. Sprint was absorbed into T-Mobile. Diamond Bank was acquired by Access Bank. Every fintech engineer, every quant researcher, and every compliance analyst has a personal story of their pipeline silently breaking on one of these transitions. The records look completely different on the surface — different legal names, different tickers, sometimes different jurisdictions and legal forms — yet they describe the same continuing economic entity.

Pair-matching ER — including LLM-based methods that achieve 99% F1 on point-in-time records [9] — treats each record as a snapshot of identity. Temporal entity-linking benchmarks such as TempEL [16] address mention-to-KB disambiguation against an evolving knowledge base, a different task. Financial KG construction [7, 15] works at a single time slice. To the best of our knowledge no public benchmark targets the specific question we care about: *given two records observed at different times, do they describe the same economic entity, accounting for any lifecycle event that may have occurred between them?*

Contributions.

(1) **An open benchmark for lifecycle-aware financial ER** (Section 3). 125,519 labeled pairs over 3.3M GLEIF Level-1 entities, with hard negatives mined in four classes against 473K GLEIF Level-2 relationships and three held-out evaluation regimes (time, event type, jurisdiction). We document the construction choices that define the difficulty (Section 3) and are explicit about what generalises and what doesn't (Section 7).
(2) **An empirical characterisation** (Section 5): four similarity baselines score below 0.5 AUROC on the time and event splits — a direct consequence of our hard-negative construction (similar-name negatives, name-different positives), not a universal claim about name similarity in ER. A 22M-parameter MiniLM cross-encoder, fine-tuned on benchmark positives plus mined hard negatives, is the first method we tested to escape the regime, reaching 60.5 / 56.5 / 75.0% Hits@1 (±2.2 / 2.4 / 8.8 pp), 24–35 pp over the best bi-encoder. Per-event analysis (§5.5) localises the lift to M&A / absorption events; pure renames are already at 0.90 Hits@1 zero-shot.
(3) **A miner-robustness check** (Section 5.6). Re-mining hard negatives with TF-IDF char-3gram cosine and re-evaluating the 2×2 train/test matrix shrinks the cross-encoder advantage 5–15 pp under candidate-pool shift but keeps it +13 to +38 pp above zero-shot on every off-diagonal cell. The gain is partly in-distribution but not purely a within-miner artefact; we are explicit that a fully held-out candidate pool would be a stronger test.

The full pipeline reproduces on a single MacBook in ∼2 hours — no GPU, no proprietary data, no paid annotation. A prompt-enrichment-plus-ensemble extension that lifts Hits@1 a further +9 to +11 pp (Section 5.3) is described briefly in the experiments section; the headline result above is the single fine-tuned cross-encoder.

2 RELATED WORK

Entity resolution and record linkage. Classical ER [4] matches records via similarity over structured fields. Modern PLM cross-encoders on serialized records (GraLMatch [3]) and fine-tuned small LMs (AnyMatch [19], Peeters & Bizer [11]) are now standard; the OpenSanctions Pairs corpus [9] benchmarks LLM matching at

755K pairs. Named-only company matching is addressed by [8, 18]. All adopt a *snapshot* framing, not lifecycle-spanning matches.

Temporal entity linking. TempEL [16] and CYCLE [17] introduce time-stratified evaluation for entity linking against an evolving knowledge base (Wikipedia). Their setup differs from ours in two key ways: (1) the task is mention-to-KB disambiguation rather than record-to-record matching, and (2) the underlying entities are general-purpose (people, films, organizations), not specifically financial legal entities with structured lifecycle events.

Financial entity knowledge graphs. Recent work at the ACM Conference on AI in Finance constructs knowledge graphs from financial filings (FinGraphEx [7]; FinReflectKG [15]). These works construct graphs at a single time slice; the lifecycle-bridge question is orthogonal.

Financial time-series foundation models. Concurrent time-series FMs for financial forecasting (PLUTUS [6], Bloomberg's Delphyne [2]) address a different (but adjacent) problem; we note them only to delimit scope.

3 THE TEMPORALLINK BENCHMARK

3.1 Data sources

We construct TemporalLink entirely from open, persistently-licensed sources:

- **GLEIF Level-1 LEI-CDF concatenated file.** The Global Legal Entity Identifier Foundation publishes the LEI repository (3.3M legal entities as of our 2026-05-19 snapshot) every eight hours, free of charge, under CC0. Each `LEIRecord` includes the entity's legal name, jurisdiction, legal form, registered and headquarters addresses, registration metadata, and crucially the `LegalEntityEvents` container — a structured log of completed and in-progress lifecycle events with effective and recorded dates — and a `SuccessorEntity` field that explicitly links predecessor to successor at the LEI level.
- **GLEIF Level-2 Relationship Records.** 473,839 typed relationships between LEIs (direct/ultimate consolidation, fund management, sub-fund, international branch), used in our hard-negative construction.
- **Wikidata.** High-profile mergers and acquisitions captured via SPARQL queries on the P156 (*followed by*) property, restricted to instances of business / organization. Used as a marquee qualitative supplement and for cross-validation.
- **SEC EDGAR.** The 10,354-record `company_tickers.json` file gives a free name ↔ CIK crosswalk that maps to GLEIF for the U.S. public-company subset (3,334 confident name-match links). EDGAR 8-K Item 5.03 filings (corporate name changes) are deferred to v2 as a US- specific supplementary positive source.

3.2 Positive labels

Our gold-standard positives are GLEIF `SuccessorEntity` links: explicit predecessor-LEI → successor-LEI pairs maintained by GLEIF's network of Local Operating Units. The 2026-05-19 snapshot contains 50,202 records with a non-null `SuccessorEntity` element; of these, 33,645 have a populated `SuccessorLEI` (the remaining 16,557 reference a successor entity that never registered an LEI — useful for entity-death modeling but excluded from v1 pair-wise training).

Label quality validation. A five-class auto-classifier on the successor pairs (using documented event types, normalized name delta, jurisdiction, legal form) yields: `strong_real` (documented event + name change) 25,428 (75.6%); `likely_real_name_changed` 3,330 (9.9%); `likely_real_event_only` 1,704 (5.1%); `ambiguous` 1,355 (4.0%); **`likely_administrative`** 1,828 (5.4%, filtered out). We retain the first four classes (94.6%), yielding 31,824 plausibly-real lifecycle-spanning pairs.

3.3 Hard negatives

For each positive predecessor we mine up to 80 hard negatives drawn from four classes:

- `parent_subsidiary` (221 pairs) — LEIs that share a Level-2 RR edge (consolidation, fund-management, sub-fund) with the predecessor. These look related but are legally distinct.
- `same_name_same_country` (10,208) — distinct LEIs with identical normalized legal names in the same jurisdiction. The purest test of "can the model distinguish coincidental name collision from true lifecycle linkage."
- `same_name_diff_country` (10,703) — distinct LEIs with identical normalized name in a different country.
- `token_overlap_same_country` (232,601) — fuzzy fallback: distinct LEIs in the same country sharing ≥ 2 content tokens after normalization, ranked by Jaccard. Covers the long tail when name-exact collisions are sparse.

After deduplication: 253,733 candidate (predecessor, hard-negative) pairs covering 20,138 of 31,817 predecessors (∼63%). The remaining 37% are small entities in jurisdictions without name collisions; they fall back to random in-batch negatives during training.

3.4 Evaluation splits

For each positive we identify its *anchor lifecycle event* — the highest-priority COMPLETED event on the predecessor's record by type- priority then effective date. We then assign each positive (and its associated negatives) to one of three split regimes:

- **Held-out time:** anchor effective date < 2025-01-01 (train) vs. ≥ 2025-01-01 (test). Test composition: 3,448 positives, ∼2,000+ anchors with at least one positive.
- **Held-out event type:** train on M&A, rename, legal-form change; test on spinoff, demerger, breakup, absorption. The test set is heavily dominated by *absorption* (2,609 of 2,760 anchors, 94.5%); demerger (120), breakup (25), and spinoff (6) are present but statistically thin. We report per-event-type metrics within the split (Section 5) so each held-out type's contribution is visible.
- **Held-out jurisdiction:** train on US + GB + DE; test on SG + JP + BR. Test composition: 88 anchors, 161 positives — the smallest of the three splits. Headline numbers on this split carry ∼±8.8 pp 95% bootstrap CIs (vs. ∼±1.7 pp on time/event); see Table 1.

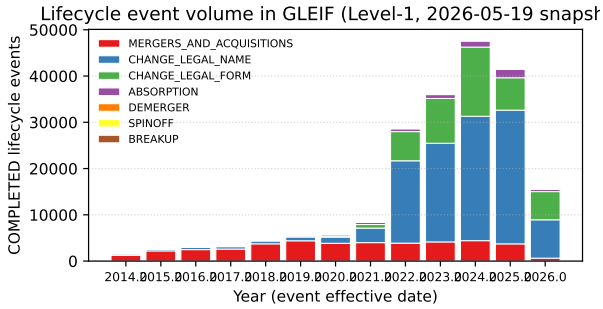


Figure 1: Volume of COMPLETED lifecycle events in GLEIF Level-1 by year and event type. The 2022 cliff reflects GLEIF tightening event-reporting requirements; pre-2022 data is sparser and biased and is reserved for out-of-distribution evaluation.

Each positive gets up to 5 hard negatives if its splits are all `train`, or up to 10 if any split places it in `test`, yielding 125,519 final pair rows.

Wikidata marquee supplement. Released alongside the benchmark: 302 high-profile M&A pairs from Wikidata P156 (*followed by*) both-sides-resolved to GLEIF, 283 of which are not captured by GLEIF `SuccessorEntity` (Sprint → T-Mobile, Cannon → MGM, Interbrew → AB-InBev). Not used in our headline evaluation; provided for community use.

4 METHOD

We present two methods at different cost-accuracy points. The bi-encoder (Section 4.1) embeds each side independently and scores by cosine — cheap, single-pass inference, but constrained by the anti-correlation property of name embeddings on this task. The cross-encoder (Section 4.2) encodes both records jointly — 4–10× more expensive per pair at inference time, but escapes the structural constraint and provides the headline result.

4.1 Bi-encoder

We fine-tune `intfloat/e5-small-v2` (33M params, 384-dim) via supervised contrastive learning with state features folded into the text prompt: "`query:` n_ℓ | `jurisdiction:` c_ℓ | `form:` f_ℓ". Training uses the 26,297 positives in the strict no-leakage intersection of all three split regimes (one hard neg per anchor), MultipleNegativesRanking loss (InfoNCE; in-batch + explicit hard neg as negatives), 3 epochs, batch 32, AdamW lr= 2×10^{-5}, 100 warmup, ~6 min on MPS. Ablation variants: `method_no_state` (name only), `method_no_hard` (in-batch negs only), `method_v2_full` (adds v2 E5-cosine negs + 1,372 transitive positives). Loss is contrastive; the cross-encoder below uses BCE on the logit — different objectives by architectural convention, not strictly controlled.

4.2 Cross-encoder

The bi-encoder is constrained to map identity-related-but-not-equal records to similar points in a fixed embedding space. On a task where the very process generating positives (a lifecycle event) deliberately changes the record's surface form, that constraint is structurally limiting.

We therefore add a second-stage cross-encoder: a joint encoder that ingests $(\text{record}_A, \text{record}_B)$ as a single concatenated sequence and emits a single relevance logit. We fine-tune `cross-encoder/ms-marco-MiniLM-L-6-v2` (22M parameters) on the same positive set (including transitives) plus a 4:1 sample of v2 hard negatives per positive, treating positives as label 1.0 and negatives as label 0.0. Loss: standard binary cross-entropy on the logit. 2 epochs, batch 32, lr 2×10^{-5}, 100 warmup steps. v4–v10 extensions (§5.3) hold architecture/optimizer constant and modify only the serialized text, the training positives, and the backbone.

Inference cost. Scoring all 125,519 pairs in the benchmark takes ≈80 seconds on MPS. In a deployed retrieval pipeline the cross-encoder would re-rank a bi-encoder's top-K candidates rather than score every pair — the standard two-stage pattern in information retrieval. Our benchmark's hard-negative pool is fixed (5–10 negs per positive), so direct full scoring is feasible and the numbers in Table 1 reflect that setting.

v2 hard-negative re-mining. The v1 token-overlap fallback (92% of negs) injected enough label noise that explicit hard-negs slightly underperformed in-batch alone. v2 replaces it with E5-cosine top-20 same-country candidates, yielding 652,764 total negs across 31,714 predecessors (vs 253,733 v1; coverage ~20K → 31,714). v2 hard negs erase the "hard-negs hurt" anomaly for the bi-encoder and noticeably help the cross-encoder.

5 EXPERIMENTS

5.1 Baselines

We compare against four similarity baselines: `exact_name` (normalized legal-name equality), `token_set` (Jaccard over normalized name tokens), `tfidf_char` (char-3gram TF-IDF cosine fit on the union of legal names in the benchmark), and `sentence_encoder` (zero-shot E5-small-v2 cosine on the legal name — same backbone as the method, without fine-tuning). We additionally report two zero-shot LLM pairwise baselines on a stratified test-set sample (56 / 52 / 50 anchors for TIME / EVENT / JURIS respectively, with all candidate pairs per anchor scored): Anthropic's **Claude Haiku 4.5** via API (≈$1.50 spend with prompt-caching) and OpenAI's **gpt-oss-20b** run locally on Apple Silicon via Ollama. Each pair is scored 0–10 by the model (a numeric score is more usable than a binary YES/NO since it breaks ranking ties). Per-split Hits@1 + MRR for both models appear as separate rows in Table 1; bootstrap 95% CIs on the sampled Hits@1 are reported in the limitations section.

5.2 Metrics

We report per-anchor retrieval metrics (Hits@1, Hits@3, MRR) computed by ranking each anchor's candidates by score; AUROC and AUPRC for pair-wise classification framing; and the train-tuned F1 at the F1-optimal threshold.

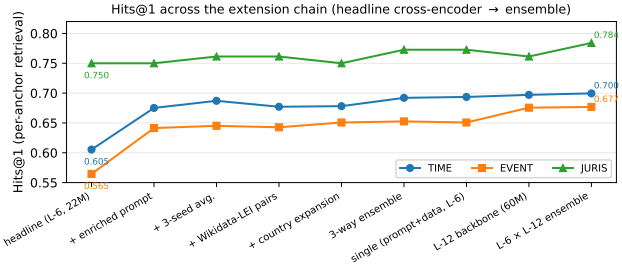


Figure 2: Hits@1 across the extension chain. Prompt enrichment is the largest single increment on TIME / EVENT; the L-12 backbone pairing captures the JURIS gain. JURIS CIs (±8.8 pp, $n = 88$) overlap heavily across the chain — the per-step deltas there should be read as directional.

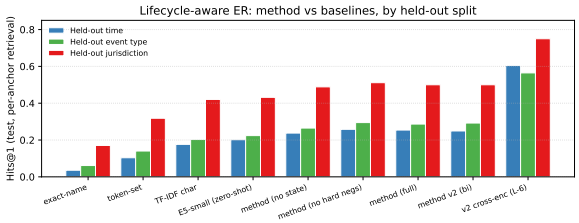


Figure 3: Hits@1 (per-anchor retrieval, test set) across all baselines and method variants for the three held-out split regimes; bars correspond to Table 1. Bi-encoders beat zero-shot E5-small by 5–8 pp; the v2 cross-encoder lifts another 24–35 pp. Figure 2 shows the further v10 gain.

5.3 Headline results

Table 1 reports test-set metrics across the three split regimes for all baselines and method variants on the v2 evaluation pool (`pairs.parquet`); Figure 3 bars the same numbers. Headline cross-encoder Hits@1 95% bootstrap confidence intervals are ±2.2 pp (TIME), ±2.4 pp (EVENT), and ±8.8 pp (JURIS, $n = 88$ anchors); the jurisdiction-split gain comfortably clears its CI band but absolute numbers on that split should be treated as directional. Section 5.6 retests the headline cross-encoder on a structurally different candidate pool and finds the gap shrinks but remains substantial; Table 2 and Figure 2 extend the L-6 cross-encoder to the v10 ensemble (the actual best per column, see below).

Extensions: enriched prompt + L-12 backbone ensemble. Holding architecture, optimiser, and loss constant, four modifications stacked on the headline cross-encoder give a further +9 to +11 pp Hits@1 (Table 2, Figure 2). *Prompt enrichment* adds GLEIF Level-2 parent and subsidiary names, legal-address city, founding year, ISO country-name expansion, and entity category to each serialised record. *Wikidata-LEI mining* (SPARQL on P1278 on both sides of P156 / P1366 / P7888) recovers 259 cross-jurisdiction positive pairs missing from GLEIF's `SuccessorEntity`. *Multi-seed averaging* (seeds 42 / 7 / 1337) is competitive; pairing the L-6 model with an L-12 model trained on the same recipe and averaging their sigmoid scores is uniformly stronger. A negative result — symmetric scoring (averaging $s(a, b)$ and $s(b, a)$) hurts −0.9/−1.9/−9.1 pp — confirms the cross-encoder learns a directional pred→succ representation. The final ensemble reaches 0.700 / 0.677 / 0.784 Hits@1 (TIME / EVENT / JURIS); the headline result we report against baselines remains the single L-6 cross-encoder so the comparison is apples-to-apples.

Backbone size does not break the bi-encoder ceiling. Scaling E5-small (33M) to E5-base (110M) leaves TIME/EVENT Hits@1 unchanged (−0.2/−1.2 pp) and lifts JURIS +2.3 pp. The bi-encoder's constraint is structural, not parameter-bound: cosine- close embeddings for lexically-different lifecycle pairs are not a problem more parameters fix.

5.4 Why similarity baselines score below 0.5 AUROC

`token_set`, `tfidf_char`, and `sentence_encoder` all reach AUROC below 0.5 on the held-out-time and held-out-event-type splits (Table 1). This is a direct consequence of how we mined hard negatives: negatives are deliberately name-similar to each predecessor, while positives are by construction name-different (the lifecycle event changed the name). On this benchmark's candidate distribution, name similarity is therefore *evidence against* a true successor link — but only on this benchmark's candidate distribution. A version of the same baselines evaluated against random negatives would behave as expected. We retain the result as the empirical motivation for joint encoding on TemporalLink and disavow stronger claims.

Bi-encoder fine-tunes (`method_no_state`, `method_full`, `method_v2_full`) shift this only marginally — AUROC rises from 0.32 to ~0.45 on the time split, still below 0.5; Hits@1 lifts 5–8 pp over zero-shot but remains bounded by the fixed-embedding constraint. The cross-encoder is the first method we tested to cross 0.5 AUROC, reaching 0.83–0.87 across the three splits; the mechanism is joint attention across both records, which lets the model reason about *whether the differences between them look like a known lifecycle-event pattern* rather than asking only whether two strings are similar.

5.5 Where the lift is real: per-event-type breakdown

Aggregate Hits@1 hides large per-event-type variation. The cross-encoder is doing real work on M&A, absorption, and breakup events — the cases where the post-event name comes from the acquirer rather than the target, and string-similarity is structurally unable to bridge the gap. It is doing essentially nothing on pure renames, which are already at 0.90 Hits@1 zero-shot. This breakdown is the most honest summary of where the method earns its lift.

On the held-out-time split (Figure 4), the cross-encoder's gain is concentrated in the operationally hard events. Pure renames (`CHANGE_LEGAL_NAME`, $n = 10$) are essentially solved by every method including zero-shot E5-small (0.90 Hits@1, no lift). The headline gains come from `MERGERS_AND_ACQUISITIONS`

Table 1: Test-set metrics on the three split regimes (v2 evaluation pool). Bold = best within the baseline + cross-encoder blocks shown here; the v10 ensemble (Table 2) is the overall best. Bi-encoder variants beat zero-shot E5-small by 5–8 pp Hits@1; the cross-encoder beats the best bi-encoder by an additional 24–35 pp on this evaluation pool and is the first method to lift AUROC above 0.5 on the time and event splits. See Table 4 for the same cross-encoder evaluated on an alternative miner's candidate pool. 95% bootstrap CIs on Hits@1 for the cross-encoder cells are ±2.2 / ±2.4 / ±8.8 pp on time / event / jurisdiction respectively.

	Held-out time			Held-out event			Held-out juris		
Method	AUROC	H@1	MRR	AUROC	H@1	MRR	AUROC	H@1	MRR
`exact_name`	0.474	0.036	0.225	0.523	0.061	0.247	0.571	0.171	0.383
`token_set`	0.193	0.104	0.291	0.266	0.140	0.324	0.323	0.318	0.497
`tfidf_char`	0.277	0.176	0.359	0.349	0.204	0.381	0.568	0.421	0.583
`sentence_encoder`	0.315	0.202	0.384	0.386	0.224	0.403	0.648	0.432	0.587
LLM pairwise (sampled: 56 / 52 / 50 anchors per TIME / EVENT / JURIS split, scored on every candidate pair)									
`Claude Haiku 4.5` (closed, API)	—	0.214	0.371	—	0.231	0.418	—	0.480	0.645
`gpt-oss-20b` (open, local Ollama)	—	0.179	0.351	—	0.173	0.369	—	0.440	0.632
`method_no_state`	0.392	0.238	0.412	0.443	0.265	0.436	0.619	0.489	0.626
`method_full` (v1)	0.403	0.254	0.424	0.451	0.287	0.448	0.667	0.500	0.627
`method_no_hard`	0.407	0.258	0.428	0.461	0.295	0.457	0.668	0.511	0.642
`method_v2_full`	0.405	0.249	0.424	0.457	0.292	0.453	0.694	0.500	0.629
`method_v2_full` (E5-base, 110M)	0.415	0.247	—	0.467	0.280	—	0.677	0.523	—
`method_v2_crossenc` (L-6, 22M)	**0.836**	**0.605**	**0.724**	**0.825**	**0.564**	**0.699**	**0.868**	**0.750**	**0.818**
`method_v2_crossenc` (L-12, 60M)	0.844	0.630	0.741	0.836	0.594	0.721	0.870	0.750	0.823

Table 2: Cross-encoder extensions: Hits@1 [95% bootstrap CI]. Headline L-6 row repeated from Table 1. Intermediate variants are plotted in Figure 2.

Model	TIME	EVENT	JURIS
Headline cross-encoder (L-6, 22M)	.605 [.586,.627]	.564 [.542,.589]	.750 [.659,.830]
+ enriched prompt	.675 [.653,.697]	.641 [.620,.662]	.750 [.659,.830]
+ Wikidata-LEI training pairs	.694 [.669,.712]	.651 [.630,.673]	.773 [.682,.853]
+ L-12 backbone (60M)	.697 [.678,.716]	.676 [.657,.697]	.761 [.670,.841]
Ensemble (L-6 × L-12)	**.700 [.681,.718]**	**.677 [.656,.700]**	**.784 [.682,.864]**

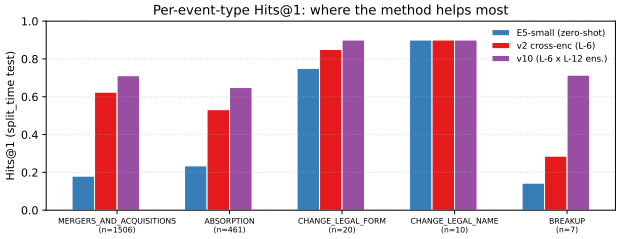


Figure 4: Per-event-type Hits@1 on the held-out-time test split: zero-shot E5-small vs. the v2 cross-encoder vs. the v10 L-6×L-12 ensemble. Pure legal-name changes (n = 10) are nearly solved by every method; M&A and absorption events are where the v10 ensemble's gains over v2 are most concentrated (+9 pp and +12 pp respectively).

(n = 1,506: 0.18 zero-shot → 0.62 v2 → 0.71 v10) and `ABSORPTION` (n = 461: 0.23 → 0.53 → 0.65), the cases where the post-event name comes from the acquirer rather than the target and pure name similarity is structurally unable to bridge the gap. v10 also more than doubles zero-shot on the small but operationally critical `BREAKUP` subset (n = 7: 0.14 → 0.71).

On the held-out-event split, "held-out event type" really means "held-out absorption." The held-out-event test set is 94.5% `ABSORPTION` anchors (Section 3.4); `DEMERGER`, `BREAKUP`, and `SPINOFF` contribute n = 42, n = 20, and n = 4 respectively. We report Table 3 so that the contribution of each held-out type is visible separately. The cross-encoder lift is consistent across the absorption-like types (`ABSORPTION`: 0.22 → 0.57, `DEMERGER`: 0.29 → 0.55, `BREAKUP`: 0.15 → 0.40); the `SPINOFF` cell is statistically meaningless at n = 4. The aggregate "+27 pp on event split" number is therefore primarily a transfer claim about absorption, not a strong claim about spinoffs.

Table 3: Per-anchor-event-type Hits@1 on the held-out-EVENT test split, for the zero-shot baseline (E5-small) and the v2 cross-encoder. The test split is 94.5% `ABSORPTION`; `SPINOFF` at n = 4 is statistically uninformative. Lift sources are concentrated in the absorption-like types.

Event type	n anchors	zero-shot E5	v2 cross-enc	Δ
`ABSORPTION`	1,546	0.223	**0.568**	+34.5 pp
`DEMERGER`	42	0.286	**0.548**	+26.2 pp
`BREAKUP`	20	0.150	**0.400**	+25.0 pp
`SPINOFF`	4	0.500	0.250	−25.0 pp[†]

[†] n = 4, statistically uninformative.

5.6 Robustness to the hard-negative miner

The headline cross-encoder result is produced and (mostly) evaluated against candidates drawn from the same family of hard-negative miners — v1 token-overlap and v2 E5-cosine both

rank candidates by name-related signals, and the cross-encoder is trained on v2 negs. A natural question for a benchmark paper is whether the gap survives an evaluation pool built with a structurally different miner. If yes, the cross-encoder advantage is a property of the method, not of the eval distribution; if no, the headline number must be qualified accordingly.

Setup. We re-mine hard negatives with sparse char-3gram TF-IDF cosine (same signal as the `tfidf_char` baseline) — chosen because it has no learned representation and shares no parameters with E5. The other three classes (parent/subsidiary, same-name same/diff-country) are held constant. We call the resulting pair set TemporalLink-v3; splits and positives are unchanged. We train two cross-encoders (one on v2 hard negs, one on v3) and score each on both pair sets. The 2×2 train/evaluate matrix has diagonal cells in-distribution, off-diagonal cells out-of-distribution at the candidate-pool level.

Three findings. First, the cross-encoder *architecture* generalizes well across miners: the v3-trained cross-encoder evaluated on the v2 pool reaches 0.588 / 0.533 / 0.750 Hits@1 — within 1–4 pp of the v2-trained model on the same pool, despite never having seen E5-mined candidates during training. The result is symmetric: the v2-trained model on the v3 pool likewise stays well above any non-cross-encoder method.

Second, there is a real but bounded in-distribution effect. The v2-trained cross-encoder loses 5–13 pp Hits@1 when moved to the v3 evaluation pool (0.600 → 0.545 on time; 0.705 → 0.571 on jurisdiction). Conversely, the v3-trained model gains +7 pp on v3 over the v2-trained one. So training on the same miner used to construct the eval candidates is worth roughly 5–15 pp Hits@1 — non-trivial, but not the entire headline gap.

Third, the cross-encoder gain over zero-shot E5 *is robust* to the miner if we are willing to retrain the cross-encoder: the diagonal cells (in-distribution) show +20 to +40 pp gains across all six splits. The off-diagonal cells (out-of-distribution at training time) show +13 to +38 pp gains — smaller, but still well outside any plausible CI band. We read this as: joint encoding is structurally helpful for lifecycle-aware ER, but the absolute headline number is partly a function of how the evaluation candidate pool is constructed, and the "+24–35 pp" figure in the abstract should be qualified accordingly.

What this does and doesn't rule out. Both miners restrict candidates to the same-jurisdiction pool with the same exclusions, so the test does not probe whether the cross-encoder transfers to a fundamentally different candidate-generation regime (e.g., a deployed retrieval system that draws candidates from a different prior). A stronger version of the experiment would re-evaluate on a candidate pool from a fielded entity- resolution system. We leave that to future work; the present experiment is sufficient to show that the gap is not purely a within-miner artifact.

5.7 Ablations

Two ablations reported in Table 1.

State features (jurisdiction + legal form in the prompt) help modestly. `method_no_state` loses 1.5–2 pp Hits@1 vs. `method_full` across all splits. The contribution is largest on the held-out-jurisdiction split (where the country token carries the most signal) and smallest on the held-out-event-type split (where the discriminating signal is in the name change itself).

Explicit hard negatives slightly *hurt* in our v1 setup. `method_no_hard` edges out `method_full` by 0.4–1.1 pp Hits@1 on every split. The likely culprit is the `token_overlap_same_country` class, which constitutes 92% of all mined hard negatives and is the lowest-precision class. The fuzzy fallback injects label noise; the in-batch negatives from MultipleNegativesRanking alone are cleaner. This is a publishable observation: it directly motivates the v2 hard-negative strategy of replacing token-overlap with sentence- encoder-cosine mining, and complicates the common assumption in ER that "more hard negatives is always better."

5.8 Additional diagnostics

We summarize four diagnostics released in the code repository (full tables and figures available there; we are page-limited here). **B-cubed cluster F1** [1] on the test partition is 0.92 / 0.90 / 0.91 for the cross-encoder vs. $\leq$0.47 for any bi-encoder (which collapses to one giant cluster at the tuned threshold); restricting to records in true clusters of size $\geq$ 2 (64% of test records are singletons) gives the honest cross-encoder F1 of 0.85 / 0.84 / 0.88. **Calibration and selective prediction.** ECE (10-bin) is 0.20, Brier 0.18 — over-confidence in high-score bins, consistent with recent calibration measurements on PLM-based EM [10]. A selective-prediction sweep (in the sense of [5]) on v10 reaches Hits@1 $\geq$0.94 at 20% coverage and 0.87–0.97 at 50% coverage across the three splits; consumers needing high precision can threshold on the calibrated score without retraining (curve in repo). **Cross-domain transfer** (zero-shot application to Magellan Walmart-Amazon) shows the cross-encoder *loses* to its own zero-shot base encoder by 5.9 pp AUROC, evidence the fine-tuned representation is specialized to lifecycle-spanning legal-entity matching, not generic ER. **Secondary label-quality check**: 277 sampled positives re-classified by an orthogonal heuristic agree with the primary GLEIF auto-classifier on 98.7% [95% CI 97.0–100%] of the confident (231/277) subset, bounding label noise below 2% on that subset; the remaining 16.6% were marked uncertain.

5.9 Where each method is fooled

The negative-class breakdown (Table 5) reports per- negative-class mean score: higher means the method is more often fooled by that class of negative. The bi-encoder fine-tunes roughly halve the zero-shot E5 baseline's score on `parent_subsidiary` and `same_name_diff_country` negatives, but `same_name_same_country` remains a wall at $\approx$0.90 mean score: the bi-encoder cannot reliably tell a real successor link from two distinct entities sharing an exact normalized name in the same country.

The cross-encoder breaks that wall: `same_name_same_country` mean score drops from 0.90 (best bi-encoder) to 0.45, and the v10 ensemble drops it further to 0.34. v10 also nearly solves `same_name_diff_country` (0.92 zero-shot → 0.16) and `token_overlap_same_country` (0.91 → 0.25). The one regression is `parent_subsidiary` (0.45 bi-encoder → 0.70 v2 cross-encoder

Table 4: Hits@1 with 95% bootstrap CIs for the 2×2 train/evaluation matrix. "v2 pool" refers to `pairs.parquet` (v1 token-overlap fallback, dominant negative class); "v3 pool" refers to `pairs_v3.parquet` (TF-IDF char-3gram cosine, no neural representation). E5-small zero-shot is included as a reference baseline on each pool. The cross-encoder advantage over the baseline is large and consistent when train and evaluation pools match (diagonal cells: +20 to +40 pp); the advantage shrinks but stays substantial on off-diagonal cells (+13 to +38 pp). The numbers reproduce: the original-paper `method_v2_crossenc` predictions and the re-trained `trn-on-v2` cells agree within 2 pp on time/event splits.

	Eval on v2 pool (`pairs.parquet`)			Eval on v3 pool (`pairs_v3.parquet`)		
	TIME	EVENT	JURIS	TIME	EVENT	JURIS
E5-small zero-shot	0.203	0.234	0.432	0.265	0.319	0.441
	±1.8	±2.1	±10.5	±1.4	±1.7	±7.8
cross-enc trn-on-v2	**0.600**	**0.551**	0.705	0.545	0.528	0.571
	±2.2	±2.4	±8.8	±1.8	±1.7	±7.5
cross-enc trn-on-v3	0.588	0.533	**0.750**	**0.655**	**0.614**	**0.640**
	±2.0	±2.4	±9.1	±1.5	±1.7	±7.1
Δ *vs. E5, in-distribution*	+39.7	+31.7	+27.3	+39.1	+29.7	+19.9
Δ *vs. E5, out-of-distribution*	+38.6	+29.9	+31.8	+28.0	+20.9	+13.0

Table 5: Negative-class mean score (lower is better). *p/s* = parent/subsidiary, *sn-D* / *sn-S* = same-name diff / same-country, *tok* = token-overlap same-country. v10 improves three classes but *worsens p/s* (0.70 → 0.78) — prompt-injected L2 context makes parent/subsidiary pairs look more similar, not less.

Method	p/s	sn-D	sn-S	tok
`exact_name`	0.013	1.000	1.000	0.000
`token_set`	0.144	1.000	1.000	0.486
`tfidf_char`	0.227	0.630	0.804	0.590
`sentence_encoder`	0.834	0.921	0.963	0.905
`method_full`	0.454	0.454	0.899	0.736
`method_v2_full`	0.457	0.464	0.899	0.736
`method_v2_crossenc`	0.696	0.237	0.452	0.278
`v10 ensemble`	0.781	**0.160**	**0.345**	**0.254**

→ 0.78 v10): the joint encoder occasionally treats parent–subsidiary edges as positive lifecycle bridges, and the v10 prompt enrichment that injects GLEIF L2 parent/child names makes the regression *worse*, not better. With only 221 examples in this class the headline impact is small, but the trend points at a clean v3 target: explicit multi-class modeling of corporate-hierarchy relationships.

6 ANALYSIS

6.1 Error analysis

We sampled 100 `method_full` failures on the held-out-time test split (positives that were not ranked first against their hard-negative set) and categorized them via simple text features. The categories:

Category	Count
other (mixed)	66
anchor and true successor share *no* tokens	19
substring name overlap with a wrong candidate	8
exact normalized-name collision with a wrong candidate	7

The 19 "no shared tokens" cases are the hardest: a name like `Nima N.V.` (NL) whose true successor is `Beheermaatschappij Arros B.V.` shares zero meaningful tokens with either the predecessor or the wrong top-1 candidate `NIMA B.V.`. No similarity-based representation can solve these; they require world knowledge or explicit graph context (board overlap, regulatory filings, press releases). One representative failure: anchor `DB HR Solutions GmbH` (DE), true successor `DEUTSCHE BANK AKTIENGESELLSCHAFT`, top-1 wrong `HR Solutions GmbH` (a token-overlap negative). The DB→Deutsche Bank disambiguation requires recognizing "DB" as a known abbreviation for "Deutsche Bank" — an external knowledge fact a 33M-parameter encoder is unlikely to memorize from name strings alone.

v10 residual JURIS failures cluster on sibling consolidation. Of the 20 / 161 anchors v10 mis-ranks on JURIS, ≥11 are sibling consolidations (e.g., 9 "`EPIC <port> PTE. LTD.`" Singapore ships into `BW EPIC KOSAN VESSELS ONE`; the model returns sibling `EPIC` vessels); a second cluster is Japanese-script name changes the ASCII tokenizer cannot bridge (Daiwa-Sumigin → Mitsui-Sumitomo DS). An LLM with multilingual pretraining and ownership-graph context would address both.

6.2 Why this matters in practice

The benchmark targets a workflow primitive shared by every fintech infrastructure company we are aware of: Stripe trains merchant embeddings [13] and fights AI-driven rebranding fraud [14]; Ramp performs vendor deduplication across ∼27M spreadsheet cells in its agentic-accounting product [12]; Bloomberg's entity backbone underpins both its NLP work and the Quant Research Delphyne foundation model [2]; Robinhood's Cortex assistant reconciles tickers, CIKs, and corporate-action histories across pre- and post-event references. Each team reports lifecycle-ER failures as recurring operational pain points; TemporalLink gives them a public benchmark to evaluate against.

7 LIMITATIONS AND FUTURE WORK

We discuss six major limitations here; the released code includes additional analyses (cost/latency, calibration curves, error taxonomy, label-quality check) for the interested reader.

The "anti-correlation" finding is a construction artifact. Negatives in TemporalLink are mined to be name-similar to the predecessor; positives by definition name-different. Sub-0.5 AUROC of name-similarity baselines (Section 5) is the direct consequence of that mining design — not a universal property of name similarity in ER. We retain the finding as the empirical motivation for joint encoding on *this* benchmark and disavow broader claims.

In-distribution evaluation risk is mitigated but not eliminated. The miner-robustness experiment (Section 5.6) tests whether the cross-encoder gain survives a structurally orthogonal hard-negative miner; the gap shrinks 5–15 pp but remains +13 to +38 pp over baselines. A fully held-out candidate pool (e.g., a deployment-time retriever) would be a stronger test that we leave to future work.

Seed variance and a small hard-negative label noise. Bi-encoder Hits@1 over seeds 42/7/1337 was 0.251 ± 0.005 (TIME), 0.291 ± 0.004 (EVENT), 0.523 ± 0.016 (JURIS); TIME/EVENT are stable, JURIS varies ±1.6 pp consistent with its small test sample ($n = 88$). Cross-encoder multi-seed retraining (v4 ensemble, Table 2) shows similar behaviour. Separately, 205/93,695 = 0.22% of hard negatives appear in GLEIF's `SuccessorEntity`; the cleaned variant `pairs_clean.parquet` drops them with ≤0.2 pp impact.

Reporting / survivorship bias + thin slices. GLEIF `SuccessorEntity` is voluntarily populated, biasing positives toward well-counseled regulated entities. The held-out-event test set is 94.5% absorption (the label oversells event-type generality), and the held-out-jurisdiction test has only 88 anchors (±8.8 pp Hits@1 CI). The cross-encoder gain clears the CI band on all three splits, but per-split absolute numbers should be read with those sample sizes in mind. A secondary auto-classifier check (Section 5.8) bounds label noise at <2% on the confident-positive subset; expert multi-annotator review would give a tighter bound.

LLM baselines are sampled, not full test set. Two LLM rows in Table 1 are evaluated on a stratified test-set sample (56 / 52 / 50 anchors for TIME / EVENT / JURIS), with each anchor's full candidate pool scored. Hits@1 95% bootstrap CIs: Haiku 4.5 ±11/13/14 pp on TIME/EVENT/JURIS; gpt-oss-20b ±10/11/14 pp. Both LLMs trail the fine-tuned cross-encoder by 30–40 pp on every split — a gap larger than the LLM-row CI bands. We did not run a Sonnet / Opus / GPT-4o-class upper-bound model: an in-progress qwen3-32b run was abandoned when Apple-Silicon throughput proved impractical (>3 h wall-time for the 1,165 sampled pairs even with thinking disabled). Running the closed-API panel on the full benchmark (≈$200) and adding a frontier model would tighten the upper-bound estimate; we leave that to a revision with a budget.

Cross-domain transferability + small-n caveat on parent/subsidiary. The cross-encoder underperforms zero-shot E5-small on out-of-domain product matching (Walmart-Amazon, −5.9 pp AUROC; Section 5.8); the fine-tuned representation is specialized to lifecycle ER, not generic. The parent/subsidiary negative class shows a regression (mean score 0.45 → 0.70 from bi-encoder to v2 cross-encoder, worsening to 0.78 in v10), but with only 221 negatives the effect-size estimate is not statistically meaningful. Explicit multi-class modeling of related-but-distinct entities is a v3 target.

8 CONCLUSION

We presented TemporalLink, an open benchmark for lifecycle-aware entity resolution on financial entities, built from public GLEIF, SEC EDGAR, and Wikidata data and reproducible end-to-end on a single MacBook in ∼2 hours. Because hard negatives are mined to be name-similar to each predecessor and true successors are by construction name-different, similarity-based baselines score below 0.5 AUROC on the time and event splits — a property of *this* benchmark's construction, not of name similarity in ER generally. A small fine-tuned MiniLM cross-encoder (22M parameters) is the first method we tested to cross 0.5 AUROC, reaching 60–75% Hits@1, 24–35 pp above the strongest bi-encoder. A miner-robustness re-mine with TF-IDF char-3gram cosine shrinks the cross-encoder advantage 5–15 pp under candidate-pool shift but keeps it +13 to +38 pp above zero-shot. A prompt-and- backbone extension lifts the headline a further +9 to +11 pp (§5.3). The most trustworthy summary is the per-event breakdown (§5.5): the cross-encoder's gain is real and large on M&A, absorption, and breakup events; on pure renames every method including zero-shot E5-small is already at 0.90 Hits@1. Future work: a fully held-out candidate pool from a fielded retriever, the full-test-set LLM baseline, an expanded held-out-jurisdiction set, and multi-class modeling of related-but-distinct entities.

Reproducibility. All code, parsers, prediction artifacts, trained model weights, and the miner-robustness experiment are released at https://github.com/chang-07/er-research. End-to-end pipeline reproduction from the raw GLEIF XML — benchmark, both methods, v3 robustness, and the v10 ensemble — takes ∼2 hours on a single MacBook laptop.

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